Vision Statement

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1 Research Vision

Mathematics without biology or biology without mathematics are not as wonderful.

Since pursuing my PhD, my research career has been highly interdisciplinary spanning analytical (e.g., mathematics, signal processing, statistics, machine learning, computer science) and neuroscience experimental (e.g., statistical analysis of responses of single cells, electrocorticography, Utah arrays, Neuropixels recordings, close-loop experimental control) subjects. I delight when I create a new method to process biological data, but I delight much further when I can apply this method to discover new aspects of brain function.

My *Statistical Neuroscience Research Group* will initially focus on creating advanced statistical methods to characterise neural and behavioral data, offline and in real time, for short- and long-duration experiments. In addition, in collaboration with experimental groups, we will apply these methods, or those created by others, to investigate behavioural and neural problems.

1.1 Latent variable models

It is challenging to extract meaning from the large number of spikes generated by high-channel count recordings probes, e.g., Neuropixels. To address this problem, my supervisor at the Gatsby Unit, Prof. Maneesh Sahani, and collaborators, created Gaussian Process Factor Analysis (GPFA Yu et al., 2009), a probabilistic method to extract low-dimensional representation (i.e., latent variables) from high-dimensional electrophysiological recordings. At the Unit I have distributed a Python implementation of a recent extension of GPFA, Sparse Variational Gaussian Process Factor Analysis (svGPFA Duncker and Sahani, 2018), which uses a sparse variational method to substantially reduces the memory and time complexity for the estimation of GPFA models and, critically, allows point-process observations, eliminating the need of binning spikes. I have improved and optimized svGPFA, and explored its capabilities to characterize several neural datasets.

In collaboration with the laboratory of Marcus Stephenson-Jones at the SWC, I have extracted svGPFA neural latents from striatal Neuropixels recordings in mice performing a stereotyped poking sequence task (Thompson et al., 2024). I then used these neural latents as inputs to a Hidden Markov Model to decode, with impressive accuracy, the mice behavioral states. Critically, this is the first position decoder that uses the spiking activity of whole populations and does not require the estimation of place fields of single units.

We are now observing that patterns of neural latents observed during behavior replay during sleep periods. If these observations are confirmed, it will be the first time that replay is identified using continuous latent variables of neural population activity, bypassing the estimation of place fields, which could open a new chapter in memory research.

In collaboration with Prof. Maneessh Sahani, my group will continue developing svGPFA and will collaborate with experimental groups at NPP and CDB to investigate the performance of svGPFA for position decoding and replay detection in the hippocampus.

1.2 Naturalistic, long-duration and continual experimentation

At the SWC and Gatsby Unit, we are pioneering a new type of naturalistic, long-duration and continual experimentation, the Aeon project, that will revolutionise neuroscience experimentation (**?**).

In 2023-2024 I mentored a very talented undergraduate student from NPP, Ms. Zimo Li, in a Simons Foundation SURFiN fellowship, where she performed the first comprehensive statistical analysis of Aeon data.

The achievements of Zimo were outstanding. She processed very complex data, that at that stage were poorly organised, and skillfully applied sophisticated statistical models (e.g., linear dynamical systems, hidden Markov models, point process regression) and produced the first statistical Aeon discoveries, which were included in the Aeon platform paper, with Zimo as coauthor. The collaboration with Zimo taught me that I can do excellent research with students that do not have a solid mathematical background.

In collaboration with the Aeon team at the Gatsby Unit and SWC, my group will continue developing statistical methods for the analysis of Aeon data. Due to the large size of Aeon recordings, conventional batch machine learning (ML) methods cannot be used to process Aeon data. Aeon requires online methods able to process non-stationary data, and are computationally very efficient to handle very large datasets. A detailed description of our ML data processing plans for Aeon appear in a collaborative grant proposal between the Gatsby Unit and the Allen Institute for Neural Dynamics we submitted to BBSRC/NSF in 2024.

1.3 Intelligent experimental control

Current neuroscience experimentation is mostly performed in open loop with predefined sets of rules designed by experimenters. More informative experiments can be designed if they could be controlled in close loop by intelligent inferences performed on behaviour and neural recordings.

Bonsai is today the most widely used software for close loop experimental control. In 2022 I realised that adding ML functionality to Bonsai could make possible a radically new type of close loop intelligent experimentation, and lead the successful application of a BBSRC proposal (BB/W019132/1) to build the Bonsai.ML package.

Most current ML methods operate offline, trained and performing inference on datasets previously recorded. However, Bonsai.ML requires a different type of ML methods, that are trained and perform inference on streams continuously delivering data in real time. We build this type of ML into Bonsai.ML. It now contains probabilistic ML methods for real time processing of behavioral (e.g., kinematics inference with linear dynamical systems, behavioral state estimation with hidden Markov models) and neural data (neural latent variables inference with linear dynamical systems, hyppocampal position decoding and replay detection with a point process state-space model). Please refer to the documentation of Bonsai.ML.

We are not integration large language models (LLMs) into Bonsai. First, we are using LLMs to help Bonsai users build correct and optimised workflows using natural language. Second, we are adding an LLM node to Bonsai to incroporate the large semantic knowledge of LLMs into the experimental controls loop. Please refer to the Open Source for the Life Science proposal we are currently preparing to fund these developments.

My group will continue developing online probabilistic models to enable intelligent experimental control in Bonsai.

1.4 Collaborations

2 Teaching Vision

References

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